

Decentralized SLM Swarm Reasoning for Constrained Planning: A Tree-of-Thought and Ant Colony Optimization Approach

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Abstract: Large Language Models (LLMs) have demonstrated strong capabilities in reasoning, planning, and tool use. However, many LLM-based reasoning systems are still deployed as centralized agents, where a single large model is responsible for interpreting the task, performing the reasoning process, and producing the final answer. This design can be computationally expensive and may be less suitable for distributed or resource-constrained environments. To explore an alternative, this paper studies decentralized collective reasoning using multiple Small Language Model (SLM)-based agents coordinated through swarm-intelligence principles. The goal is not to replace centralized LLM systems outright, but to investigate whether lightweight SLM agents can collectively approximate high-quality solutions to constrained planning problems through structured, distributed coordination. Our framework is inspired by the pheromone-based reasoning approach of Chari et al. (Chari et al., 2025), but is adapted to the specific demands of calendar scheduling and implemented entirely with SLMs. Concretely, it combines a fixed five-level Tree-of-Thought (ToT) decomposition with an Ant Colony Optimization (ACO)-style branch-selection mechanism and a deterministic verifier that scores candidate schedules against task constraints. We validate this framework on 400 curated instances from the NaturalPlanner calendar scheduling benchmark, using the dataset as a controlled testbed for assessing whether swarm-based SLM reasoning can improve planning performance. Across three settings, a single-call SLM baseline achieves a solve rate of 0.2475, a multi-agent ToT configuration without ACO reaches 0.3950, and the full ToT+ACO framework achieves the best result at 0.4000. These findings show that structured multi-agent ToT search provides the main performance gain over direct single-agent solving, while the additional contribution of ACO is modest in the present setup. At the same time, the fact that pheromone-guided search achieves the strongest overall result suggests that its benefit may become clearer under larger-scale experiments with broader search budgets and longer iterative adaptation. As such, this work serves both as a task-specific adaptation of pheromone-based reasoning to calendar scheduling and as an initial empirical foundation for further study of SLM-based swarm planning.

1 INTRODUCTION

Large Language Models (LLMs) have shown strong performance in reasoning, planning, decision-making, and tool-use tasks. Their ability to interpret natural language instructions, reason over constraints, and generate structured outputs has made them useful for a wide range of complex problem-solving applications. In particular, planning tasks require a model to understand task requirements, identify possible actions, evaluate constraints, and select a valid solution. These capabilities make language models promising candidates for solving natural lan-

guage planning problems.

Despite these strengths, most language-model-based reasoning systems are still designed around centralized execution. In such systems, a single large model is usually responsible for understanding the complete task, performing the reasoning process, and generating the final answer. While this approach can be effective, it also introduces several limitations. Large centralized models often require substantial computational resources, may increase inference cost, and can be difficult to deploy in distributed or resource-constrained environments. They may also create a single point of failure, because the quality

of the final solution depends heavily on one model’s reasoning trajectory. These limitations motivate the exploration of alternative architectures in which reasoning is distributed across multiple smaller agents.

In parallel, research in multi-agent systems and swarm intelligence has shown that groups of agents can solve complex problems through local interaction, cooperation, and iterative improvement. Swarm intelligence is inspired by collective behaviour in natural systems such as ant colonies, bird flocks, and fish schools. In such systems, individual agents follow relatively simple local rules, but their interactions can produce useful global behaviour. This idea is particularly relevant for optimisation and planning problems, where exact solutions may be computationally expensive or infeasible. Instead of relying on a single central decision-maker, swarm-based methods use distributed exploration and reinforcement to identify good approximate solutions.

Motivated by these observations, this work investigates whether the semantic reasoning capabilities of Small Language Models (SLMs) can be combined with swarm intelligence mechanisms to support decentralised planning. The central idea is to examine whether multiple SLM-based agents can collectively reason over constrained planning tasks and improve solution quality through structured exploration and indirect coordination. In this setting, each agent contributes one reasoning trajectory, while swarm-inspired mechanisms guide the collective search toward more promising solution paths.

To implement this idea, the proposed framework combines Tree-of-Thought (ToT) reasoning (Yao et al., 2023) with an Ant Colony Optimisation (ACO)-inspired coordination mechanism (Dorigo, 1992; Dorigo et al., 1996). The framework is inspired by the pheromone-based reasoning approach of Chari et al. (Chari et al., 2025), but is adapted specifically for the calendar scheduling setting studied in this work. The framework is inspired by the pheromone-based reasoning approach of Chari et al., but is adapted specifically for the calendar scheduling setting studied in this work. In particular, the adaptation introduces a domain-specific ToT structure tailored to scheduling subtasks, restricts the reasoning framework to SLM-based agents only, and uses deterministic task-specific quality scoring rather than a generic evaluation signal. Tree-of-Thought reasoning allows the system to explore multiple intermediate reasoning paths instead of producing only a single direct answer. This is useful for planning tasks because different reasoning paths may represent different interpretations of constraints, different candidate solutions, or different strategies for satisfying

task requirements. The ACO-inspired component introduces a swarm-based reinforcement mechanism, where reasoning branches associated with higher-quality solutions receive stronger reinforcement and become more likely to be selected in later iterations.

The framework is evaluated using the calendar scheduling task from the NATURAL PLAN benchmark (Zheng et al., 2024). Calendar scheduling is a constrained planning problem in which the system must identify a valid meeting time while considering participant availability, working-hour limits, meeting duration, allowed days, and possible preferences. This task is suitable for evaluating collective reasoning because a generated answer is only correct if all relevant constraints are satisfied simultaneously. A model may fail by overlooking a participant’s busy interval, selecting a slot with the wrong duration, violating the allowed time window, or ignoring a preference. Therefore, the task provides a meaningful test case for studying whether structured multi-agent reasoning can improve solution quality.

The objective of this work is not to claim that decentralised SLM-based systems can replace centralized LLMs or exact optimisation methods. Rather, the goal is to explore whether lightweight multi-agent reasoning can provide an effective and resource-aware alternative for specific constrained planning tasks. By studying the relationship between single-agent reasoning, multi-agent candidate exploration, and swarm-inspired reinforcement, this paper contributes to the broader research direction of collective semantic intelligence and decentralised AI-based planning.

Here are the main contributions of this paper are :

- A task-specific adaptation of an existing pheromone-guided reasoning framework is introduced to study whether multiple SLM agents can collaboratively solve constrained planning problems.
- An empirical comparison is conducted across three settings: a single-agent baseline, a multi-agent ToT-only configuration, and a full ToT+ACO configuration.
- The results show that structured multi-agent ToT reasoning clearly improves over direct single-agent solving, while the ACO component, although not yet conclusively validated in the current setup, achieves the best overall result and suggests promising potential for larger-scale future study.

2 Scope and Research Objectives

The scope of this work is exploratory and methodological. The study focuses on investigating whether decentralized coordination principles from swarm intelligence can be combined with the semantic reasoning capabilities of language models to support collective problem solving in constrained planning tasks.

The main research objectives of this work are as follows:

- to design and evaluate a decentralized multi-agent reasoning framework in which multiple small language model agents cooperate to solve constrained planning tasks;
- to integrate swarm intelligence mechanisms that support coordination, solution refinement, and collective decision-making without central control;
- to evaluate the proposed approach using existing benchmark tasks for complex planning, with a specific focus on a curated calendar scheduling subset of the NaturalPlanner benchmark.

This work does not aim to replace centralized LLM-based reasoning systems. Instead, it explores an alternative system design in which multiple smaller reasoning agents cooperate through decentralized coordination. Such an approach may be beneficial in application contexts where robustness, distributed reasoning, interpretability of intermediate reasoning steps, or iterative solution improvement are important.

3 Related Work

1. Architectures of Current Multi-Agent LLM Frameworks

Over the past few years, a number of multi-agent LLM frameworks have been developed, each taking a different approach to coordinating agents. A broad survey of these systems groups them by how agents communicate, divide work, and coordinate across tasks (Guo et al., 2024). Some frameworks, such as AutoGen, set up conversational loops where specialized agents exchange messages back and forth to solve a task (Wu et al., 2024). Others, such as MetaGPT, use fixed Standard Operating Procedures (SOPs) to break work into an assembly-line of steps (Hong et al., 2024). CAMEL explores communicative role-playing agents for studying cooperative behaviour in LLM-based societies (Li et al., 2023). The

most widely discussed problem is context explosion. In a system where every agent talks to every other agent, the number of messages grows roughly as $O(N^2)$ with the number of agents. This means that as the team gets bigger, agents start running out of space in their context window just to keep track of the conversation. A related problem is that most of these frameworks rely on a central "Manager" or "Architect" agent to assign tasks and keep things on track. If that node fails, times out, or produces a hallucination, the whole pipeline stalls. This is a classic single point of failure. Finally, frameworks like CAMEL and MetaGPT tend to be rigid. The roles and workflows are defined in advance, which makes them hard to adapt when a task takes an unexpected turn. True swarm intelligence requires agents to react to local conditions, and fixed pipelines are not designed for that (Guo et al., 2024).

2. Decentralized Coordination and Evolutionary Swarms

A newer wave of research is trying to move away from central orchestrators. Instead of one agent directing the others, these systems let agents organize themselves based on local information and direct peer-to-peer interaction. AgentNet structures the agent network as a Directed Acyclic Graph (DAG), where tasks are passed directly between agents based on their evolving specialization (Yang et al., 2025). This removes the bottleneck of a central coordinator and means the system can keep running even if one agent fails.

SwarmAgentic takes a different approach. It draws on Particle Swarm Optimization (PSO) to maintain and evolve a population of agent system configurations at once (Zhang et al., 2025). Rather than manually designing agent roles, the system generates and refines entire agent teams from scratch through a language-driven search process. This is a significant step toward systems that can build themselves.

ReEvo shows that LLMs can act as hyper-heuristics. The key idea is that an LLM can not only apply coordination rules but also reflect on them and propose improvements through an evolutionary loop (Ye et al., 2024). ReEvo has been shown to solve combinatorial optimization problems in this way, without manual rule engineering. This is relevant here because a similar reflective search process could be used to evolve coordination strategies in a multi-agent system.

One limitation of these systems is that they have mostly been tested on narrow benchmarks. The performance of population-based search depends

heavily on how the fitness function is defined, and it is not yet clear how well these methods generalize to open-ended tasks where success is hard to measure.

3. Integrating Swarm Intelligence (SI) for Planning and Reasoning

Swarm intelligence draws on decades of biology and computer science research. The core insight is that simple local rules, followed by many agents simultaneously, can produce intelligent collective behaviour without any central controller. Dorigo established much of the theoretical foundation for Ant Colony Optimization in the early 1990s (Dorigo, 1992), later formalized in the Ant System framework (Dorigo et al., 1996). Reynolds introduced local flocking rules for simulated agents through the Boids model (Reynolds, 1987), and Kennedy and Eberhart introduced Particle Swarm Optimization (Kennedy and Eberhart, 1995). Artificial Bee Colony optimization later provided another population-based search mechanism inspired by bee foraging behaviour (Karaboga and Basturk, 2007). These foundational ideas are now being adapted for LLM-based systems, though the translation is not always straightforward.

Ant Colony Optimization (ACO) is especially relevant for this work. In biological ant colonies, ants find short paths to food by leaving and following pheromone trails. In computational ACO, this idea is translated into an optimization process where high-quality paths are reinforced and weaker paths gradually evaporate (Dorigo, 1992; Dorigo et al., 1996). ACO-ToT applies the same idea to reasoning itself: LLM-based agents explore a tree of possible reasoning steps, and successful chains of logic are reinforced with pheromones, allowing the swarm to converge on better answers over time (Chari et al., 2025). Early results suggest that ACO-guided reasoning can outperform standard Chain-of-Thought prompting (Wei et al., 2022) and extend Tree-of-Thought search (Yao et al., 2023), though broader comparison across diverse benchmarks is still needed. Boids and robot swarms represent another relevant line of work. Reynolds’s Boids model shows that three simple rules, separation, alignment, and cohesion, are enough to produce realistic flocking behaviour (Reynolds, 1987). LLM2Swarm applies this thinking to robot swarms, where physical robots use LLMs to reason, plan, and coordinate locally without a central server (Strobel et al., 2024). Each robot interprets its neighbours’ states and adjusts its be-

haviour accordingly. This is one of the first systems to test bio-inspired swarm coordination with real hardware, and it highlights both the promise and the engineering difficulty of the approach. Artificial Bee Colony (ABC) optimization is also relevant for configuration search. The ABC algorithm divides a population of search agents into three roles: employed bees exploit known good solutions, onlooker bees select among them probabilistically, and scout bees explore new areas when current solutions stagnate (Karaboga and Basturk, 2007). Applied to LLM systems, this type of meta-heuristic can optimize the configuration of an agent team itself, searching over prompt templates, retrieval settings, and agent counts. The connection to Automatic Prompt Engineer (APE) is relevant here: if LLMs can automatically generate and evaluate prompts (Zhou et al., 2023), then ABC-style search gives a structured way to explore the space of possible configurations systematically. A key open question is whether these SI principles transfer cleanly to discrete token-based systems. Biological pheromones are continuous signals; LLM “pheromones” are discrete annotations or updated memory entries. How well the analogy holds under real conditions is something that needs empirical testing, not just theoretical argument.

4. Pheromone-based Learning of Optimal Reasoning Paths

A closely related line of work is the pheromone-guided reasoning framework of Chari et al., who propose Ant Colony Optimization-guided Tree of Thought (ACO-ToT) as a method for improving language-model reasoning through iterative path exploration and reinforcement (Chari et al., 2025). Rather than treating reasoning as a single linear chain, their approach constructs a centralized Tree-of-Thought structure and uses multiple specialized LLM “ants” to traverse candidate reasoning paths, with branch selection influenced by both heuristic preferences and accumulated pheromone strength. Reasoning paths are then evaluated and reinforced across iterations, allowing the system to progressively favor more productive trajectories. The authors position this method as a biologically inspired alternative to standard Chain-of-Thought prompting (Wei et al., 2022) and Tree-of-Thought prompting (Yao et al., 2023), motivated by the need for more efficient search over large reasoning spaces; they report gains on general reasoning benchmarks such as GSM8K, ARC-Challenge, and MATH. For the present project, this work is the closest concep-

tual precursor, since it introduces the core idea of pheromone-guided reasoning-path optimization. However, our framework adapts that idea to a different setting: constrained calendar scheduling, SLM-only agents, a domain-specific ToT decomposition, and deterministic task-specific quality scoring rather than the original generalized reasoning setup.

5. Benchmarking and Performance Gaps

One of the clearest pieces of evidence motivating this research is the TravelPlanner benchmark (Xie et al., 2024). TravelPlanner tests how well language agents can produce valid, multi-constraint travel plans given a set of requirements. TravelPlanner tests how well language agents can produce valid, multi-constraint travel plans given a set of requirements. The results are striking: GPT-4 achieves a success rate of only 0.6% on this challenging benchmark, where the agent must satisfy all constraints simultaneously. This is not a niche failure case. It reveals a broader gap between what current LLMs can do in isolation and what a reliable planning system needs to deliver. The evaluation gap is worth naming clearly. Benchmarks like TravelPlanner measure end-to-end plan validity, not just fluency or approximate correctness. A plan that violates one constraint fails completely, even if every other detail is right. This kind of hard evaluation exposes weaknesses that softer benchmarks miss. It also makes the case for swarm-based coordination: if a single large model fails this badly, perhaps a team of smaller specialized agents, coordinating through stigmergic signals and local interaction rules, can do better by collectively covering more of the constraint space. A complementary benchmark in this direction is NATURAL PLAN, which evaluates natural-language planning across three task types: Trip Planning, Meeting Planning, and Calendar Scheduling (Zheng et al., 2024). Unlike a fully interactive tool-use benchmark, NATURAL PLAN provides models with task-relevant context derived from tools such as Google Flights, Google Maps, and Google Calendar, allowing planning performance to be assessed directly from natural-language inputs and outputs. The three tasks span different aspects of constrained planning: trip planning emphasizes itinerary construction under travel and temporal constraints, meeting planning focuses on coordinating meetings under availability and travel-time considerations, and calendar scheduling requires selecting valid meeting slots given participants’ schedules and timing constraints. In

this paper, we use the calendar scheduling component of NATURAL PLAN as a focused evaluation domain for testing whether decentralized SLM-based swarm reasoning improves constraint satisfaction over direct single-agent solving. It should be noted that most existing swarm-based LLM systems have not been evaluated on standardized planning benchmarks at all, and comparison remains difficult because different papers use different datasets and metrics. This work contributes a benchmark-grounded study that places a task-adapted ToT+ACO swarm framework, a ToT-only variant, and a single-agent baseline under the same constrained planning setting, enabling a clearer assessment of what structured multi-agent reasoning actually adds.

4 Method & Approach

4.1 System Instantiation

The current implementation instantiates the proposed swarm framework on a constrained calendar scheduling task drawn from the calendar scheduling subset of the NATURAL PLAN benchmark (Zheng et al., 2024). In each task, the system must identify a meeting slot that satisfies participant availability, allowed day constraints, working-hour limits, and the required meeting duration. Some tasks also include soft preferences, such as earliest availability or avoiding a specific time period. The final output is returned in a fixed format: Here is the proposed time: <Day>, <HH:MM> - <HH:MM> . This setting is useful for the present study because it frames planning as a constrained search problem in which correctness depends on satisfying multiple interacting constraints simultaneously.

Each dataset item is provided as a few-shot prompt containing solved examples followed by the active task instance. During generation, the full 5-shot prompt is provided to the agents so that the examples remain available as in-context guidance. For task-feature construction, the implementation derives a task-specific representation from the active instance using the current-task text together with dataset metadata. For scoring, the verifier uses the zero-shot prompt when it is available; otherwise, it uses the final extracted task block from the few-shot prompt. This separation allows the examples to support reasoning while keeping evaluation tied only to the active task.

4.2 Decentralized Multi-Agent Architecture

The implemented system is inspired by the pheromone-guided reasoning framework proposed by Chari et al. in *Pheromone-based Learning of Optimal Reasoning Paths* (Chari et al., 2025), but adapts that idea in several important ways for the present study. While the original framework uses a parent LLM to dynamically construct a task-specific Tree-of-Thought structure and later solve the task using the best reasoning path found across iterations, our implementation focuses specifically on constrained planning tasks and modifies the framework to operate with SLM agents only, a fixed domain-specific ToT structure for calendar scheduling, and a deterministic task-specific scoring mechanism. In addition, rather than handing the final best path back to a parent model for answer generation, the current system selects the best result from the final iteration itself as the final prediction. This choice is consistent with the logic of pheromone-based learning, since the later iterations reflect the most updated branch-selection preferences after repeated reinforcement and evaporation, and therefore represent the most informed stage of the search process for the current task.

The implemented system consists of multiple autonomous language-model agents operating on a shared planning instance without a central manager. Each agent performs local reasoning independently and contributes one candidate solution trace. Coordination is achieved indirectly through a shared pheromone state rather than through explicit exchange of full reasoning histories. This design follows the principle of stigmergic coordination, where collective behavior emerges through modifications to a shared environment rather than through top-down control.

Centralized multi-agent language-model frameworks often rely on manager agents, dense conversational loops, or fixed orchestration pipelines, all of which can introduce context growth, coordination bottlenecks, and single points of failure (Guo et al., 2024). In contrast, the present system retains a common reasoning structure across agents while allowing branch-level diversity and task-local adaptation through swarm-inspired reinforcement.

4.3 Tree-of-Thought Search Structure

The reasoning process is implemented as a fixed five-level Tree-of-Thought (ToT) graph, following the idea of structured intermediate reasoning search (Yao et al., 2023), illustrated in Figure 1. Each level corre-

sponds to a distinct planning subproblem and contains three alternative reasoning branches, yielding a structured but diverse search space. The five levels are:

1. task parsing,
2. calendar organization,
3. feasible-slot generation,
4. preference application and slot selection,
5. final verification and answer formatting.

At the first level, the branches represent different ways of interpreting the task: one emphasizes extracting explicit scheduling constraints first, another begins by identifying the participants and their individual commitments, and a third focuses first on any stated preferences before integrating the remaining constraints. At the second level, the alternatives reflect different ways of organizing the calendar information, such as structuring it participant by participant, day by day, or through a mixed normalization strategy that combines both views. At the third level, the branches differ in how candidate meeting windows are generated: one computes free intervals directly and intersects them, another enumerates possible slots and filters them, and a third progressively removes conflicts from a broader candidate window. At the fourth level, the reasoning branches capture alternative selection policies, such as preferring the earliest valid slot, prioritizing explicit soft preferences, or choosing a more conservative slot that avoids borderline conflicts. At the fifth level, the alternatives vary in how the final answer is produced, ranging from direct answer generation to lightweight self-checking and explicit validation of output format and slot consistency.

The structure is directly encoded in the implementation as a graph of sequential reasoning nodes. At each node, the agent receives the original prompt together with the intermediate outputs accumulated from previous levels and is instructed to generate only the intermediate result required for the current branch. After the fifth level, a separate finalization step produces the final answer in canonical format. This design constrains the model to reason stage by stage, while also making the full planning trajectory explicit and observable, which is useful both for analysis and for branch-level optimization.

4.4 Agent Execution Procedure

For each task, the system schedules n agent runs during each search iteration. Every agent traverses the five-level ToT graph once and therefore produces one complete reasoning trace, consisting of the selected

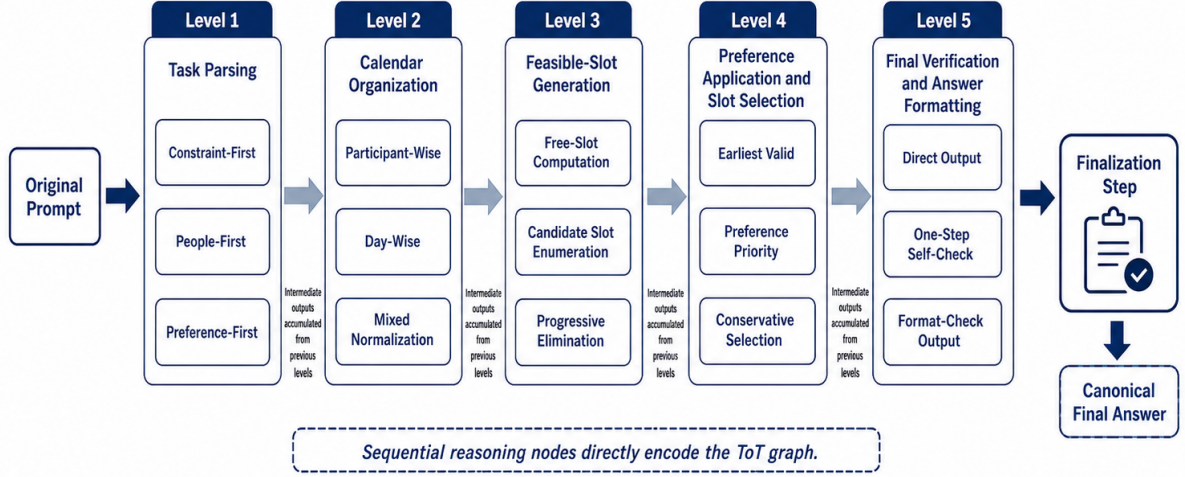


Figure 1: Tree-of-Thought Structure

branch path, intermediate reasoning outputs, final answer, cumulative path score, and a deterministic quality score. The path score is computed as the product of the branch-selection probabilities encountered along the trace, while the quality score is assigned by the task-specific verifier based on how well the final answer satisfies the scheduling constraints.

Within an individual agent, execution is sequential across the five ToT levels. Across agents, execution is concurrent, although the runtime can cap the number of in-flight agents to reduce burst throttling from the model provider. This keeps the search space controlled while still allowing different agents to explore different reasoning paths. In effect, each SLM agent serves as a local reasoning module embedded in a structured search process rather than as an unconstrained planner.

4.5 ACO-Based Branch Selection

Branch selection at each ToT level is governed by Ant Colony Optimization (ACO), a swarm-intelligence method based on pheromone-guided reinforcement of promising paths (Dorigo, 1992; Dorigo et al., 1996). For level i and branch j , the probability of selecting that branch is defined as

$$P_{ij} = \frac{(\tau_{ij} + \epsilon)^\alpha (\eta_{ij} + \epsilon)^\beta}{\sum_k (\tau_{ik} + \epsilon)^\alpha (\eta_{ik} + \epsilon)^\beta} \quad (1)$$

where τ_{ij} denotes the pheromone value of branch j at level i , η_{ij} denotes its heuristic desirability, α controls the influence of pheromone, β controls the influence of heuristic bias, and ϵ is a smoothing constant used to avoid zero-probability branches. When ACO is disabled, the same ToT structure is preserved, but branches are sampled uniformly at random. This

provides a non-ACO baseline under the same search budget.

The heuristic desirability term η_{ij} is implemented through lightweight task-aware rules derived from features available before search begins, including the number of participants, the number of valid days, meeting duration, the presence of soft preferences, and whether the task requests the earliest available slot. At the early reasoning levels, the heuristic favors branches that foreground the most structurally important information for the given task: tasks with many participants are biased toward reasoning patterns that organize information around individual attendees, while multi-day or longer-duration tasks favor branches that emphasize constraint extraction and day-based normalization. During feasible-slot generation, simpler tasks are biased toward direct computation of shared free intervals, whereas larger or more combinatorial tasks favor strategies that either enumerate candidate windows more broadly or progressively eliminate conflicts step by step. At the selection stage, tasks with explicit earliest-availability requirements or other soft preferences receive corresponding bias toward branches that prioritize those criteria, while tighter tasks receive a mild preference for more conservative choice strategies. This heuristic design is intentionally simple but well motivated: it does not attempt to solve the task in advance, but it steers early exploration toward reasoning styles that are more likely to match the structural demands of the instance, thereby giving pheromone learning a more informative starting distribution from which to adapt.

4.6 Local Pheromone Update

The current implementation uses local pheromone adaptation within each task. At the start of every task, the pheromone table is initialized uniformly. After each search iteration, the pheromone values are updated according to

$$\tau_{ij} \leftarrow (1 - \rho)\tau_{ij} + \sum_k \Delta_{ij}^{(k)} \quad (2)$$

where

$$\Delta_{ij}^{(k)} = \begin{cases} Q_k, & \text{if branch } (i, j) \text{ is part of agent } k\text{'s path,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Here, Q_k denotes the quality score assigned to the final prediction of agent k . Consequently, branches that appear in higher-quality traces receive larger deposits, while older influence decays through evaporation. This mechanism implements the exploration-exploitation balance: early iterations encourage path diversity, whereas later iterations increasingly favor branches associated with stronger candidate solutions.

An important design decision is that pheromone memory is local to a single task. The system does not carry the pheromone state across the entire dataset in the main solve workflow. Instead, the swarm adapts within one instance, produces a final decision, and then resets before the next task. This keeps the methodology focused on instance-level decentralized coordination rather than on global cross-task policy learning.

4.7 Deterministic Constraint-Based Scoring

Candidate solutions are scored using a deterministic task-based verifier rather than an additional LLM judge. This deterministic constraint-based scoring mechanism is one of the key task-specific adaptations in our framework, as it is tightly coupled to the structure of calendar scheduling and designed to evaluate outputs directly against scheduling constraints. The verifier parses the proposed time from the final answer and evaluates it against the active task. Five hard checks are applied: (1) whether the selected day is permitted, (2) whether the duration is correct, (3) whether the slot lies within the allowed work window, (4) whether it overlaps any participant’s busy interval, and (5) whether it violates any hard preference.

To support preference-sensitive scoring, the verifier enumerates all hard-valid meeting slots in 30-minute increments across the allowed day set and

sorts them chronologically. This enables the framework to determine not only whether a proposed slot is feasible, but also whether it is the earliest feasible option and how well it satisfies soft preferences.

For hard-valid candidates, the quality score is computed as

$$Q = 0.5 + 0.5 r_{\text{soft}} \quad (4)$$

where r_{soft} is the soft-preference satisfaction ratio. For hard-invalid candidates, the score is computed as

$$Q = 0.49 r_{\text{hard}} \quad (5)$$

where r_{hard} is the fraction of hard checks passed. This scoring rule enforces a strict ranking separation between feasible and infeasible answers: any hard-valid candidate scores above any hard-invalid one. The verifier therefore functions both as an evaluation mechanism and as the reward source for pheromone reinforcement. This is particularly important in planning tasks, where fluent text alone is not a meaningful indicator of correctness (Xie et al., 2024).

4.8 Per-Instance ACO Workflow

The main experimental methodology follows a per-instance, classical ACO-style workflow. For a given task, the system first loads the few-shot prompt, extracts the active task for scoring, builds the task-feature representation, and initializes a fresh local pheromone table. It then performs T search iterations. In each iteration, n agents are launched, each agent traverses the five ToT levels, each candidate answer is scored by the deterministic verifier, and, when ACO is enabled, the pheromone state is updated using the scored traces.

At the end of the search, the framework selects the best trace from the final iteration only. It does not select the globally best trace observed across all iterations. In the current workflow, the earlier iterations are treated as adaptation phases through which the swarm reshapes its branch preferences, while the final iteration represents the stabilized coordinated search state from which the official prediction is drawn. This workflow is the main evaluation mode for the implemented system because it preserves task independence, confines pheromone learning to the current instance, and allows ACO-enabled and ACO-disabled runs to be compared under the same total budget of $n \times T$ agent traversals. The overall workflow of this per-instance ACO procedure is illustrated in Figure 2.

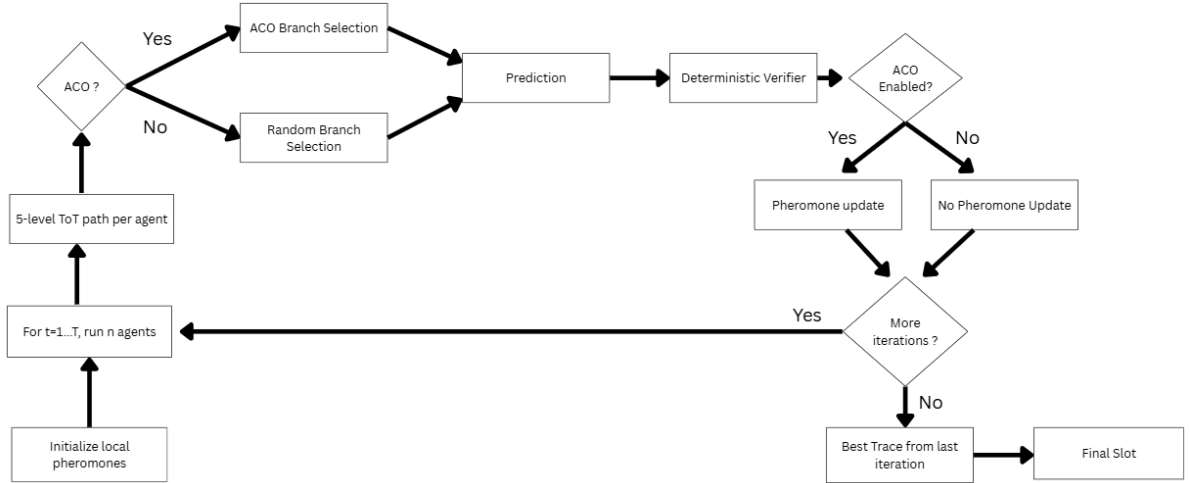


Figure 2: ACO and ToT Approach

4.9 Logging and Reproducibility

The implementation records both prediction outputs and structured execution metadata. In addition to the dataset-level output, the system writes structured JSONL run logs in solve mode. Each task-level log entry stores the `original_task_prompt`, the `scoring_task_prompt`, the final prediction, the metadata summary, and the returned traces. The logged traces correspond to the final iteration and include selected branches, intermediate reasoning outputs, path information, final answers, and quality annotations.

5 Experiments

5.1 Experiment Overview

The experiment compares three configurations: a single-call SLM baseline, a multi-agent multi-iteration Tree-of-Thought (ToT) framework without Ant Colony Optimization (ACO), and a multi-agent ToT framework enhanced with ACO-based branch selection. This comparison is designed to isolate two questions: first, whether structured decentralized multi-agent reasoning improves performance over direct single-agent solving; and second, whether pheromone-guided coordination provides additional benefit beyond ToT-based exploration alone.

The calendar scheduling task is a suitable test case for this research because it requires more than fluent text generation. The model must parse participant availability, identify calendar conflicts, respect working-hour and day constraints, account for meet-

ing duration, and handle optional soft preferences before producing a valid meeting time in the required format. This makes the task a constrained planning problem in which failure can arise at multiple stages of reasoning. A model may correctly identify the duration but overlook a participant conflict, or it may find a feasible slot while failing to satisfy a stated preference. The task therefore provides a meaningful setting for evaluating whether structured reasoning and decentralized candidate exploration improve solution quality.

The central idea of the experiment is that the semantic reasoning ability of language models can be combined with swarm-intelligence principles to support collective planning. In the implemented framework, the fixed five-level ToT structure decomposes reasoning into multiple candidate paths, while the ACO component introduces iterative pheromone-based reinforcement over branch choices within each task. The expectation is that ToT improves the breadth and structure of reasoning, while ACO may support iterative optimisation by strengthening candidate paths that receive higher quality scores.

5.2 Dataset

The experiments use the calendar scheduling subset of the NATURAL PLAN benchmark (Zheng et al., 2024). The original dataset contains 1,000 instances, evenly divided across ten task categories: 2-person scheduling tasks spanning 1 to 5 days, and 3-, 4-, 5-, 6-, and 7-person scheduling tasks each defined over a single day, with 100 samples in each category.

For this study, a curated subset of 400 instances was constructed by taking the first 40 samples from

Table 1: Description of Calendar Scheduling dataset

Field	Description
num_people	Number of participants who must attend the meeting.
num_days	Number of valid candidate days over which the meeting may be scheduled.
duration	Required duration of the meeting.
prompt_5shot	The 5-shot prompt used for generation, containing solved examples followed by the active scheduling task.
golden_plan	Ground-truth reference solution provided by the benchmark.
prompt_0shot	Zero-shot version of the active task, used for task-specific evaluation and scoring without reliance on the in-context examples.
pred_5shot_pro	Prediction field retained from the original benchmark structure.
pred_model	Model identifier associated with the stored prediction field in the dataset format.

each category. This reduction was motivated by practical cost constraints while still preserving diversity in task size and difficulty, allowing meaningful evaluation across varying numbers of participants and planning horizons.

Each JSON item contains the task metadata and prompt material required for generation and evaluation. These fields capture the core scheduling parameters, prompt variants, reference solution, and stored benchmark prediction metadata. The meanings of the main fields are summarized in Table 1.

5.3 Evaluation Metric

The primary evaluation metric used for the NaturalPlanner calendar scheduling task is solve rate. This metric measures the proportion of task instances for which the predicted meeting time exactly matches the benchmark reference solution. For each response, the evaluation extracts the proposed day, start time, and end time from the model output and compares them directly against the corresponding ground-truth slot. A task is counted as solved only when all three components match exactly; otherwise, it is marked as incorrect.

Let $\hat{x}_i = (d_i^{\text{pred}}, s_i^{\text{pred}}, e_i^{\text{pred}})$ denote the predicted slot for instance i , where d_i^{pred} , s_i^{pred} and e_i^{pred} denote the predicted day, start time and end time respectively. Similarly, let $x_i = (d_i^{\text{gold}}, s_i^{\text{gold}}, e_i^{\text{gold}})$ denote the corresponding ground-truth slot, where d_i^{gold} , s_i^{gold} , and e_i^{gold} represent the reference day, start time, and end time, respectively. Formally, solve rate is defined as

$$\text{Solve Rate} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{x}_i = x_i], \quad (6)$$

where N is the total number of evaluated instances and $\mathbf{1}[\cdot]$ is an indicator function that returns 1 when the predicted slot exactly matches the reference slot

and 0 otherwise. This metric provides a strict end-to-end measure of planning correctness, since even a small deviation in the predicted day, start time, or end time causes the prediction to be counted as a failure.

6 Results and Analysis

6.1 Overall Solve Rate Comparison

The overall solve rates for the three evaluated configurations are shown in Table 2.

Table 2: Overall solve rate comparison across evaluated configurations.

Configuration	Solve Rate
Single SLM baseline	0.2475
Multi-agent ToT without ACO	0.3950
Multi-agent ToT with ACO	0.4000

The single SLM baseline achieved an overall solve rate of 0.2475 across 400 tasks. This indicates that the single-agent model solved approximately one quarter of the evaluated calendar scheduling problems. In comparison, the multi-agent Tree-of-Thought framework without ACO achieved an overall solve rate of 0.3950. The full multi-agent ToT+ACO framework achieved the highest overall solve rate, with a value of 0.4000.

The improvement from the single SLM baseline to the ToT+ACO framework is substantial. The solve rate increases from 0.2475 to 0.4000, which corresponds to an absolute improvement of 0.1525, or 15.25 percentage points. In relative terms, the ToT+ACO framework improves performance by approximately 61.6% over the single SLM baseline. This result indicates that the proposed multi-agent reasoning framework is considerably more effective

than direct single-agent solving on the curated benchmark.

The improvement from the single SLM baseline to the multi-agent ToT-only configuration is also large. The ToT-only framework increases the solve rate from 0.2475 to 0.3950, corresponding to an absolute improvement of 14.75 percentage points. This shows that most of the observed gain comes from the introduction of multi-agent Tree-of-Thought reasoning. In other words, the main performance improvement appears to result from generating, exploring, and evaluating multiple candidate reasoning paths rather than relying on a single reasoning trajectory.

The additional improvement from ToT-only to ToT+ACO is smaller. The solve rate increases from 0.3950 to 0.4000, corresponding to an absolute improvement of 0.5 percentage points. This means that, in the current experimental setup, the incremental benefit of ACO over ToT alone remains limited and not yet conclusive. At the same time, the fact that the ACO-enhanced configuration achieves the best overall result suggests that pheromone-guided search retains meaningful promise. Its contribution may become more visible under conditions that better support adaptive search, such as greater reasoning-tree diversity or depth, a larger number of iterations, and repeated evaluation across multiple random seeds for both the ToT-only and ToT+ACO settings. From this perspective, the present results should be seen not as a rejection of ACO, but as an initial indication that its effect likely depends on a richer and more extensively sampled search regime.

6.2 Category-Level Performance

To better understand the behaviour of the evaluated methods, the results were also analysed across different task categories. The benchmark contains tasks with varying numbers of participants and planning horizons, with 40 samples in each category. The category-level solve rates are shown in Table 3.

The category-level results broadly reinforce the main overall finding: both multi-agent methods outperform the single-SLM baseline in most categories. This advantage is particularly visible in one-day scheduling tasks with several participants, where the reasoning problem requires combining multiple availability constraints within a limited time window. For instance, in the 3-people, 1-day category, performance rises from 0.250 for the single-SLM baseline to 0.550 for ToT-only and 0.575 for ToT+ACO. A similarly strong pattern appears in the 5-people, 1-day category, where the solve rate increases from 0.175 to 0.300 for ToT-only and to 0.525 for ToT+ACO.

Table 3: Category-level solve rates across evaluated configurations.

Category	Single SLM	ToT-only	ToT+ACO
2 people, 1 day	0.40	0.60	0.575
2 people, 2 days	0.25	0.45	0.325
2 people, 3 days	0.10	0.325	0.35
2 people, 4 days	0.15	0.20	0.125
2 people, 5 days	0.075	0.10	0.175
3 people, 1 day	0.25	0.55	0.575
4 people, 1 day	0.375	0.375	0.35
5 people, 1 day	0.175	0.30	0.525
6 people, 1 day	0.325	0.525	0.525
7 people, 1 day	0.375	0.525	0.475

These results suggest that decentralized candidate exploration is especially helpful when several participant constraints must be jointly satisfied.

The gains for larger one-day tasks remain notable even when ACO does not add a further improvement. In the 6-people, 1-day category, both ToT-only and ToT+ACO achieve a solve rate of 0.525, clearly above the single-agent baseline of 0.325. Likewise, in the 7-people, 1-day category, both multi-agent settings outperform the baseline, although ToT-only slightly exceeds ToT+ACO. Taken together, these cases suggest that the strongest and most consistent benefit comes from structured multi-agent Tree-of-Thought reasoning itself.

For multi-day tasks with two participants, the pattern is more mixed. In some categories, the ACO-enhanced system performs best, such as the 2-people, 3-days and 2-people, 5-days categories, where it slightly exceeds ToT-only while remaining clearly above the single-agent baseline. In other categories, however, ToT-only performs better, most notably in the 2-people, 2-days and 2-people, 4-days settings. This indicates that the effect of ACO is not yet stable across category types and may depend on properties such as search depth, branching diversity, and the amount of iterative exploration available within each task.

Overall, the category-level analysis supports two main conclusions. First, multi-agent Tree-of-Thought reasoning provides a robust improvement over direct single-agent solving across most task categories. Second, the additional effect of ACO is promising but uneven: it strengthens performance in some settings, particularly certain multi-day or medium-complexity cases, but does not yet produce a uniformly superior category-level pattern. This again suggests that further experimentation is needed before making strong claims about the independent contribution of pheromone-guided coordination.

6.3 Interpretation of the Improvement over the Single SLM Baseline

The improvement over the single-SLM baseline can be understood as a shift from direct single-path generation to structured multi-path reasoning. In the baseline setting, the model produces one final answer in a single attempt, so performance depends entirely on whether it correctly parses all constraints and carries out the full scheduling computation without error. Calendar scheduling is especially sensitive to such failures, because a prediction is only correct if it simultaneously satisfies day, duration, work-hour, availability, and preference constraints; even a small mistake makes the final answer invalid.

The Tree-of-Thought framework reduces this fragility by replacing one-shot generation with structured exploration of multiple reasoning trajectories. Different paths can reflect different interpretations of the task, different candidate slots, or different strategies for satisfying constraints. The multi-agent setting strengthens this further by allowing several agents to explore the task in parallel, reducing dependence on any single reasoning attempt and increasing the chance that at least one agent identifies a valid solution. In this way, the framework operationalises the central idea of collective semantic search through decentralized reasoning.

The results are consistent with this interpretation. The solve rate rises from 0.2475 for the single-SLM baseline to 0.3950 for the multi-agent ToT-only framework, indicating that much of the observed gain comes from the structured multi-agent reasoning process itself. This suggests that, for constrained planning tasks such as calendar scheduling, the main benefit lies in moving from one-shot answer generation to iterative exploration over multiple candidate solutions.

6.4 Interpretation of the ACO Contribution

The role of ACO in this framework is to introduce a swarm-inspired optimization mechanism over reasoning paths. In classical Ant Colony Optimization, candidate paths that lead to better solutions are reinforced through pheromone updates and become more likely to be selected in later exploration. In the present setting, this idea is adapted to reasoning search: candidate trajectories that receive higher deterministic quality scores are reinforced, allowing later iterations to benefit from earlier successful exploration. In this sense, ACO adds a form of distributed memory to the framework and turns multi-agent reasoning into more

than simple parallel sampling.

In the current results, the overall solve rate increases from 0.3950 for ToT-only to 0.4000 for ToT+ACO. This is a positive difference, but it is small and should be interpreted cautiously. Under the present experimental conditions, the results are not strong enough to support a definitive claim that ACO consistently improves performance beyond Tree-of-Thought reasoning alone.

Several factors may explain why the ACO effect is not yet clearly visible. First, the dataset remains relatively small, with 400 tasks overall and 40 samples per category, so modest changes in solved counts can noticeably affect the reported solve rates. Second, the current reasoning tree may still be too limited for pheromone-based optimization to fully express its advantage, since ACO is most effective when search spaces are large enough to support repeated reinforcement of useful paths. Third, the search budget of 8 agents and 6 iterations per task, while sufficient for demonstrating feasibility, may be too small for stable pheromone convergence. Finally, the current final-iteration-best selection policy may understate ACO's benefit, because useful paths discovered earlier in the search are not directly preserved unless they remain competitive in the final iteration.

Taken together, these results suggest that ACO is promising but not yet conclusively validated in the present setup. The small improvement shows that pheromone-guided coordination can be integrated into an SLM-based reasoning framework, but further experiments with larger search spaces, deeper or more diverse reasoning trees, longer iterative runs, and alternative selection policies are needed to determine whether ACO delivers robust gains beyond ToT alone.

6.5 Limitations of the Current Results

The current results should be interpreted with several limitations in mind. First, although 400 tasks are sufficient for a preliminary study, the evaluation set remains relatively small for drawing strong generalizable conclusions, especially because it is divided into ten categories with only 40 samples each. As a result, some category-level differences may be sensitive to a small number of successes or failures.

Second, the reported results appear to be based on a single main run for each configuration. Without repeated runs across different random seeds or sampling conditions, it is difficult to determine how much of the observed difference reflects stable behaviour rather than run-specific variation. This is particularly relevant for multi-agent and iterative reasoning systems, where small changes in candidate generation can af-

fect final outcomes.

Third, the current reasoning tree and search budget are still limited. Tree-of-Thought methods benefit from richer path diversity, and ACO is most effective when there are enough candidate paths and reinforcement cycles to clearly distinguish strong reasoning trajectories from weaker ones. The present setup may therefore understate the potential of both structured search and pheromone-guided adaptation.

Fourth, the evaluation uses a curated subset rather than the full NaturalPlanner benchmark. While this keeps the experiment manageable and controlled, it may not fully reflect the broader diversity of planning tasks in the complete dataset. The results should therefore be interpreted as evidence of feasibility and initial improvement, rather than as a definitive demonstration of general planning superiority.

Finally, solve rate alone does not capture all aspects of reasoning quality. A failure may result from several different causes, including incorrect parsing, faulty constraint intersection, preference violation, formatting mismatch, or final answer mismatch. More detailed error analysis would be valuable for understanding which types of failures are reduced by ToT and which remain challenging even under multi-agent coordination.

6.6 Summary of Findings

The results provide preliminary empirical support for the central idea of the project. The multi-agent reasoning frameworks substantially outperform the single SLM baseline on the curated calendar scheduling benchmark. The strongest improvement is observed when moving from direct single-agent solving to multi-agent Tree-of-Thought reasoning. This suggests that structured candidate exploration and collective reasoning are effective for constrained planning tasks.

The ToT+ACO framework achieves the highest overall solve rate, reaching 0.4000 compared with 0.2475 for the single SLM baseline. This corresponds to a clear improvement over the baseline and supports the argument that language-model-based agents can benefit from collective reasoning structures. However, the additional improvement over ToT-only is small, increasing from 0.3950 to 0.4000. Therefore, the current experiment supports the value of ToT more strongly than the independent contribution of ACO.

The ACO component remains theoretically meaningful because it introduces swarm-inspired coordination, reinforcement, and collective memory into the reasoning process. However, its empirical effect is not yet strongly visible in the current setup. This is

likely due to the limited sample size, small category-level subsets, limited reasoning-tree scale, and relatively small number of iterations. Future experiments should therefore evaluate the framework on larger datasets, repeat runs multiple times, increase the breadth and depth of the reasoning tree, and compare different selection policies.

In conclusion, the results show that combining semantic reasoning with collective search mechanisms can improve planning performance over a single SLM baseline. The findings are consistent with the broader research direction of collective semantic intelligence, where multiple language-model agents cooperate through structured reasoning and swarm-inspired coordination. Although the empirical contribution of ACO is not yet conclusively established in the current setup, the fact that the ToT+ACO configuration achieves the best overall result suggests that pheromone-guided coordination has meaningful potential. The present findings therefore position ACO not as a disproven component, but as a promising mechanism whose value is likely to become clearer under larger-scale and more extensively sampled experimental settings.

7 CONCLUSION

This paper investigated whether swarm intelligence mechanisms can be combined with Small Language Model (SLM)-based agents to support decentralised reasoning for constrained planning tasks. The main motivation was to explore an alternative to single, centralised language-model reasoning systems, especially in settings where computational efficiency, scalability, and distributed problem solving are important. To study this direction, the work proposed a multi-agent framework that combines Tree-of-Thought reasoning with an Ant Colony Optimization-inspired coordination mechanism.

The proposed system was evaluated on a curated subset of 400 calendar scheduling tasks from the NaturalPlanner benchmark. The results show that the multi-agent reasoning configurations clearly outperform the single SLM baseline. The single SLM baseline achieved a solve rate of 0.2475, while the multi-agent Tree-of-Thought framework without ACO achieved 0.3950. The full ToT+ACO framework achieved the highest overall solve rate of 0.4000. This corresponds to an absolute improvement of 15.25 percentage points over the single-agent baseline. These findings indicate that structured multi-path reasoning and collective candidate generation can improve performance on constrained planning

tasks.

The strongest improvement was observed when moving from direct single-agent solving to multi-agent Tree-of-Thought reasoning. This suggests that the main benefit of the proposed approach comes from decomposing the reasoning process into multiple stages and allowing several agents to explore alternative reasoning trajectories. In calendar scheduling tasks, where the final answer must satisfy several hard constraints simultaneously, this type of structured exploration reduces the fragility of one-shot generation and increases the chance of finding a valid solution.

The contribution of ACO was positive but limited in the current experimental setup. The ToT+ACO framework slightly outperformed the ToT-only framework, but the difference was small. Therefore, the current results support the value of Tree-of-Thought-based multi-agent reasoning more strongly than the independent effect of ACO. Nevertheless, the ACO component remains important from a methodological perspective because it introduces swarm-inspired reinforcement, local adaptation, and collective memory into the reasoning process. Further experimentation is required to determine whether stronger ACO effects emerge with larger datasets, deeper reasoning trees, more iterations, or alternative selection strategies.

Overall, this work provides preliminary evidence that decentralised SLM-based agents can collectively approximate solutions for complex planning problems more effectively than a single SLM baseline. The findings support the broader research direction of collective semantic intelligence, where multiple language-model agents cooperate through structured reasoning and swarm-inspired coordination. At the same time, the results show that this is still an exploratory approach, and stronger empirical validation is needed before making general claims about robustness, scalability, or superiority over centralised systems.

8 Future Scope

The present work provides an initial exploration of decentralized SLM-based reasoning with Tree-of-Thought and Ant Colony Optimization for calendar scheduling, but several directions remain open for future work.

First, the framework should be evaluated on the full NaturalPlanner calendar scheduling dataset of 1,000 instances rather than the curated 400-instance subset used here. A larger and more diverse evaluation set would support stronger conclusions, reduce sensitivity to category-level fluctuations, and provide

a more reliable comparison across configurations.

Second, future experiments should include repeated runs with different random seeds. Since multi-agent reasoning involves stochastic candidate generation and search behaviour, both the ToT-only and ToT+ACO configurations should be evaluated multiple times under the same budget, with average solve rates and standard deviations reported to make comparisons more statistically meaningful.

Third, the ACO component should be tested under a broader range of settings. Future work should examine whether stronger effects emerge with more agents, more iterations, or richer reasoning trees, and should also compare alternative selection policies such as global-best selection across all iterations rather than selecting only from the final iteration.

Fourth, the framework should be evaluated with multiple Small Language Models. Different SLMs may vary in reasoning quality, constraint handling, instruction following, and formatting reliability. Comparing models such as Phi, Gemma, Mistral, and Llama-based variants under the same baseline, ToT-only, and ToT+ACO settings would provide a clearer picture of model-dependent behaviour and enable a comparative leaderboard.

Fifth, future work should include more detailed error analysis. While solve rate is useful as an overall metric, it does not reveal whether failures arise from parsing errors, incorrect duration calculation, faulty constraint intersection, preference violations, or formatting problems. A finer-grained analysis would clarify which error types are reduced by structured multi-agent reasoning and which remain difficult.

Finally, the framework should be tested beyond calendar scheduling. Although this task is a useful constrained planning benchmark, it represents only one planning domain. Evaluating the approach on other benchmarks, such as TravelPlanner (Xie et al., 2024) or the other NATURAL PLAN task types (Zheng et al., 2024), would help determine whether decentralized SLM-based swarm reasoning generalizes more broadly.

Overall, future work should focus on scaling the dataset, repeating experiments across multiple runs, comparing several SLMs, strengthening the ACO evaluation, and expanding to additional planning tasks. These steps would provide a stronger empirical basis for assessing whether swarm-coordinated SLM agents can become a practical resource-aware alternative to single-agent reasoning systems.

REFERENCES

- Chari, A., Tiwari, A., Lian, R., Reddy, S., and Zhou, B. (2025). Pheromone-based learning of optimal reasoning paths. *arXiv preprint arXiv:2501.19278*.
- Dorigo, M. (1992). *Optimization, Learning and Natural Algorithms*. PhD thesis, Politecnico di Milano, Milan, Italy.
- Dorigo, M., Maniezzo, V., and Colorni, A. (1996). Ant system: Optimization by a colony of cooperating agents. *IEEE Transactions on Systems, Man, and Cybernetics, Part B: Cybernetics*, 26(1):29–41.
- Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., Wiest, O., and Zhang, X. (2024). Large language model based multi-agents: A survey of progress and challenges. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 8048–8057.
- Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., and Schmidhuber, J. (2024). Metagpt: Meta programming for a multi-agent collaborative framework. In *International Conference on Learning Representations*.
- Karaboga, D. and Basturk, B. (2007). A powerful and efficient algorithm for numerical function optimization: Artificial bee colony algorithm. *Journal of Global Optimization*, 39(3):459–471.
- Kennedy, J. and Eberhart, R. C. (1995). Particle swarm optimization. In *Proceedings of the IEEE International Conference on Neural Networks*, pages 1942–1948.
- Li, G., Hammoud, H. A. A. K., Itani, H., Khizbullin, D., and Ghanem, B. (2023). CAMEL: Communicative agents for “mind” exploration of large language model society. In *Advances in Neural Information Processing Systems*.
- Reynolds, C. W. (1987). Flocks, herds, and schools: A distributed behavioral model. In *Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH ’87*, pages 25–34.
- Strobel, V., Dorigo, M., and Fritz, M. (2024). LLM2Swarm: Robot swarms that responsively reason, plan, and collaborate through llms. *arXiv preprint arXiv:2410.11387*.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., and Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35.
- Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A. H., White, R. W., Burger, D., and Wang, C. (2024). Autogen: Enabling next-gen llm applications via multi-agent conversation framework. In *Proceedings of the First Conference on Language Modeling*.
- Xie, J., Zhang, K., Chen, J., Zhu, T., Lou, R., Tian, Y., Xiao, Y., and Su, Y. (2024). Travelplanner: A benchmark for real-world planning with language agents. *arXiv preprint arXiv:2402.01622*.
- Yang, Y., Chai, H., Shao, S., Song, Y., Qi, S., Rui, R., and Zhang, W. (2025). Agentnet: Decentralized evolutionary coordination for llm-based multi-agent systems. *arXiv preprint arXiv:2504.00587*.
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., and Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36.
- Ye, H., Wang, J., Cao, Z., Berto, F., Hua, C., Kim, H., Park, J., and Song, G. (2024). Reevo: Large language models as hyper-heuristics with reflective evolution. *arXiv preprint arXiv:2402.01145*.
- Zhang, Y., Lin, C., Tang, S., Chen, H., Zhou, S., Ma, Y., and Tresp, V. (2025). Swarmagentic: Towards fully automated agentic system generation via swarm intelligence. *arXiv preprint arXiv:2506.15672*.
- Zheng, H. S., Mishra, S., Zhang, H., Chen, X., Chen, M., Nova, A., Hou, L., Cheng, H.-T., Le, Q. V., Chi, E. H., and Zhou, D. (2024). NATURAL PLAN: Benchmarking llms on natural language planning. *arXiv preprint arXiv:2406.04520*.
- Zhou, Y., Muresanu, A. I., Han, Z., Paster, K., Pitis, S., Chan, H., and Ba, J. (2023). Large language models are human-level prompt engineers. In *International Conference on Learning Representations*.